

Postulate 1 of QM

Wavefunction postulate

Recap of complex linear vector space

① State space
Consider $|v\rangle \in V$

(i) for all $(\forall)$
 $\lambda \in \mathbb{C}$,

$$\lambda |v\rangle \in V$$

(ii) If $|v_1\rangle, |v_2\rangle \in V$

then $\alpha |v_1\rangle + \beta |v_2\rangle \in V$

② Notation
Bra-ket notation

$$|v\rangle \leftrightarrow \langle v|$$

ket space bra space

③

Superposition principle

④ dimension & basis vectors
Matrix representation

In a d-dim vector space
 $|v\rangle$ is a d-dimensional
"column vector" with
complex entries

$$|v\rangle = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_d \end{pmatrix} \in \mathbb{C}^d$$

$$\langle v| = (v_1^* \ v_2^* \ \dots \ v_d^*) \in \mathbb{C}^{1 \times d}$$

$$\textcircled{a} \textcircled{ii} \alpha |v_1\rangle + \beta |v_2\rangle = \alpha v_1 + \beta v_2$$

$$\begin{pmatrix} \alpha v_{11} + \beta v_{21} \\ \alpha v_{12} + \beta v_{22} \\ \vdots \\ \alpha v_{1d} + \beta v_{2d} \end{pmatrix}$$

$$\begin{pmatrix} \alpha v_{11} + \beta v_{21} \\ \alpha v_{12} + \beta v_{22} \\ \vdots \\ \alpha v_{1d} + \beta v_{2d} \end{pmatrix} = \begin{pmatrix} \alpha v_{11} + \beta v_{21} \\ \alpha v_{12} + \beta v_{22} \\ \vdots \\ \alpha v_{1d} + \beta v_{2d} \end{pmatrix}$$

⑤ Inner product $\langle v|w\rangle$

$\langle v|w\rangle$ row vector $\times$ column vector
matrix multiplication
bra ket

$$\begin{pmatrix} v_1^* & v_2^* & \dots & v_d^* \end{pmatrix} \cdot \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{pmatrix}$$

$$= v_1^* w_1 + v_2^* w_2 + \dots + v_d^* w_d \in \mathbb{C}$$

$$\begin{aligned} \textcircled{a} \textcircled{ii} \quad \alpha |v_1\rangle + \beta |v_2\rangle &\leftrightarrow \alpha \langle v_1| + \beta \langle v_2| \\ &= \alpha \langle v_1| + \beta \langle v_2| \\ &= \langle \alpha v_1 + \beta v_2| \end{aligned}$$

$$\textcircled{a} \textcircled{ii} \quad \lambda |v\rangle \leftrightarrow \lambda \langle v|$$

ⓐ Outer product

Ket bra $\xleftrightarrow{\text{matrix multiplication}}$ column vector $\times$ row vector

ⓐ $|v\rangle\langle w|$
ket bra

$$= \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_d \end{pmatrix} \cdot (w_1 \ w_2 \ \dots \ w_d)$$

$d \times 1$ $1 \times d$

Operators
← Square matrices

$$\leftarrow = \begin{pmatrix} v_1 w_1 & v_1 w_2 & \dots & v_1 w_d \\ v_2 w_1 & v_2 w_2 & \dots & v_2 w_d \\ \vdots & \vdots & \ddots & \vdots \\ v_d w_1 & v_d w_2 & \dots & v_d w_d \end{pmatrix}$$

$d \times d$

Postulates of QM.

(1) Wavefunction of a physical system is a vector in a complex linear vector space
→ Hilbert space

and the vector is of

unit norm

where norm is defined as square of length of the vector.
Here norm of $|\psi\rangle = \langle\psi|\psi\rangle$

Consider a general $|\psi\rangle$ in the d -dimensional vector space with basis vectors given by

$$|1\rangle, |2\rangle, \dots, |d\rangle$$

$$\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{d \times 1}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}_{d \times 1}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}_{d \times 1}$$

Then $|\psi\rangle = c_1|1\rangle + c_2|2\rangle + \dots + c_d|d\rangle = \boxed{\sum_{i=1}^d |c_i|^2 = 1}$ $zz^* = |z|^2$

Therefore
norm of $|\psi\rangle$
 $= \langle \psi | \psi \rangle$

$$= \begin{pmatrix} c_1^* & c_2^* & \dots & c_d^* \end{pmatrix}_{1 \times d} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_d \end{pmatrix}_{d \times 1}$$

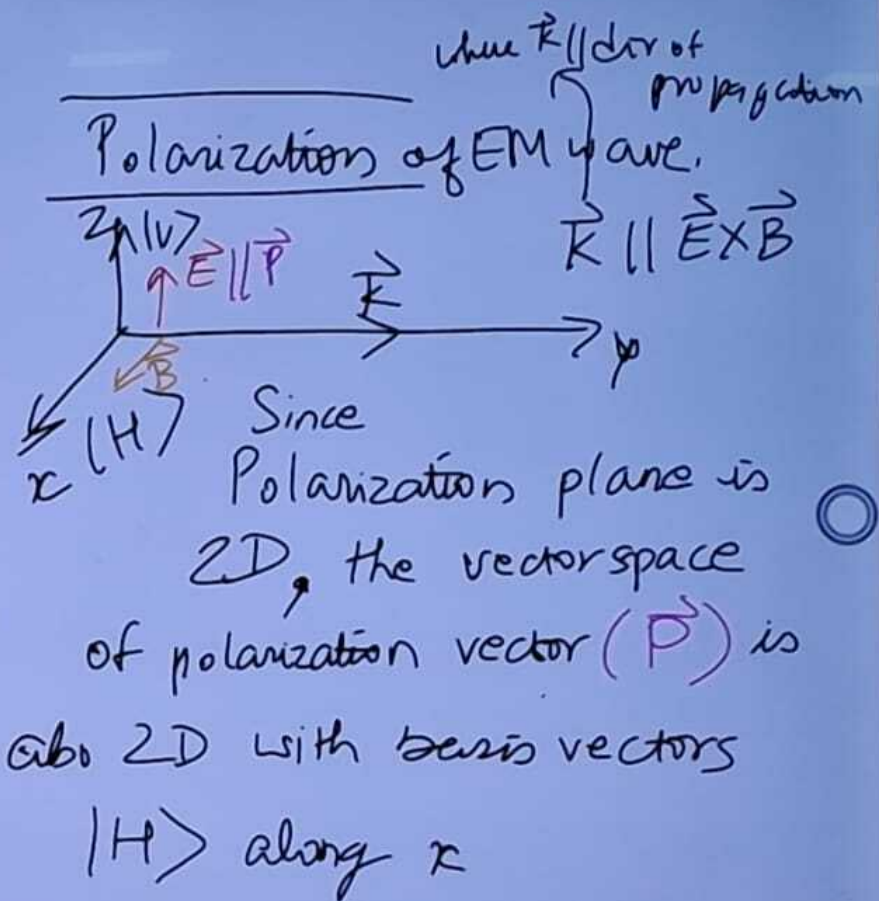
$$= c_1^* c_1 + c_2^* c_2 + \dots + c_d^* c_d$$

$$= |c_1|^2 + |c_2|^2 + \dots + |c_d|^2$$

$$= |z|^2$$

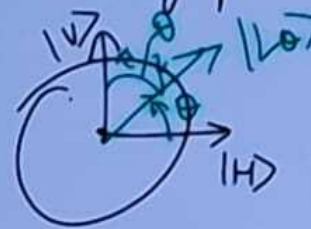
Physical systems

- Restrict to $d=2$
 - Polarization of light
 - Spin $\frac{1}{2}$
 - Two-level atom (Superconducting qubit)



So any general $|P\rangle = \alpha |H\rangle + \beta |V\rangle$

Linearly polarized



$$|L_0\rangle = \alpha |H\rangle + \beta |V\rangle$$

To find α , take

inner product with $\langle H|$

on both sides of eqⁿ (o.b.s.)

$$\begin{aligned}\langle H|L_0\rangle &= \langle H|\alpha|H\rangle + \langle H|\beta|V\rangle \\ &= \alpha \langle H|H\rangle + \beta \langle H|V\rangle\end{aligned}$$

$$\langle H|L_0\rangle = \alpha$$

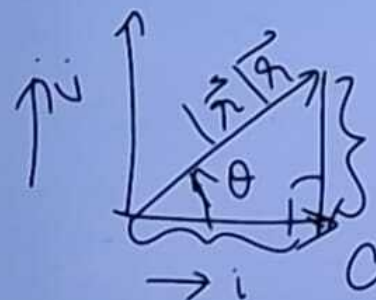
$\equiv \cos\theta$

Similarly,

$$\langle V|L_0\rangle = \beta$$

$\equiv \sin\theta$

$$|L_0\rangle = \cos\theta |H\rangle + \sin\theta |V\rangle$$



$$\vec{r} \cdot \hat{i} = (|\vec{r}| \cos\theta \hat{i} + |\vec{r}| \sin\theta \hat{j}) \cdot \hat{i} = |\vec{r}| \cos\theta$$

$$\cos\theta = \frac{\vec{r} \cdot \hat{i}}{|\vec{r}|} \Rightarrow ? = |\vec{r}| \cos\theta$$

Matrix representation

$$- |H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}_{2 \times 1} \quad |V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}_{2 \times 1}$$

$$\Rightarrow |L_\theta\rangle = \cos\theta \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin\theta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|L_\theta\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

$$\text{Check: } \langle L_\theta | L_\theta \rangle = 1 \Leftrightarrow (\cos\theta \quad \sin\theta)_{1 \times 2} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}_{2 \times 1} = C_\theta^2 + S_\theta^2 = 1$$

HW.

- Circularly polarized
- Elliptically polarized

$$|C_+\rangle, |C_-\rangle$$

$$|E_+\rangle, |E_-\rangle$$

Two level physical systems

1. Polarization states of light
2. Spin $\frac{1}{2}$ system
3. Superconducting qubit

Physical systems

— Restrict to 2 level system

$$|E_{\pm}^+\rangle = \begin{pmatrix} a \cos \theta \\ \pm j b \sin \theta \end{pmatrix}$$

Polarization states

Spin $\frac{1}{2}$ system

— Superconductivity qubit
(2 level atom)

$$a^2 \cos^2 \theta + b^2 \sin^2 \theta = 1$$

$$= a^2 \cos^2 \theta + b^2 \sin^2 \theta = 1$$

Check $\langle C_{\theta}^- | C_{\theta}^+ \rangle =$

$$\langle \cos \theta | H \pm j \sin \theta | V \rangle$$



$$\langle \cos \theta | H \pm j \sin \theta | V \rangle = \cos \theta \langle H | + j \sin \theta \langle V |$$



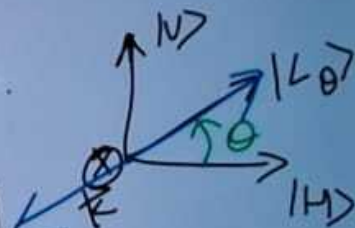
$$|0\rangle \leftrightarrow |H\rangle \quad |1\rangle \leftrightarrow |V\rangle$$

$$|L_{\theta}\rangle = \cos \theta |H\rangle + \sin \theta |V\rangle$$

$$\begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

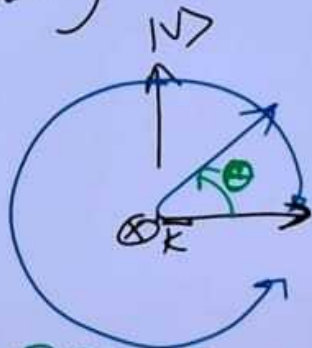
Specific cases

$$|L_{\pm \frac{\pi}{4}}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$



$$|C_{\theta}^+\rangle = \cos \theta |H\rangle + \sin \theta e^{j\frac{\pi}{2}} |V\rangle$$

$$= \cos \theta |H\rangle \pm j \sin \theta |V\rangle$$



$$|H\rangle |C_{\theta}^+\rangle = \begin{pmatrix} \cos \theta \\ \pm j \sin \theta \end{pmatrix}$$

Specific cases

$$|C_{\frac{\pi}{4}}^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ +j \end{pmatrix}$$

$$x^2 + y^2 = 1 \quad (a \cos \theta, b \sin \theta)$$

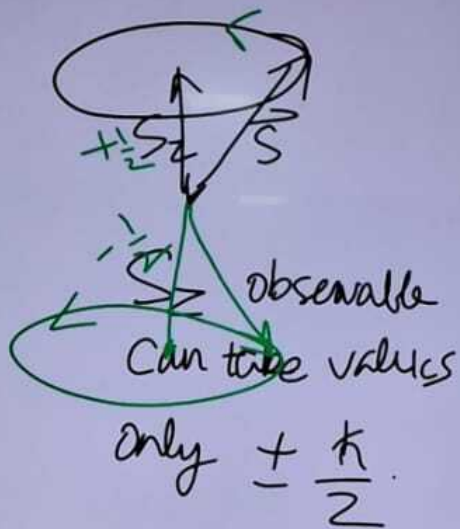
⑥ Spin- $\frac{1}{2}$

Electronic system.

$$|\uparrow\rangle, |\downarrow\rangle$$

$$|\frac{1}{2}\rangle, |-\frac{1}{2}\rangle$$

$$|\frac{\hbar}{2}\rangle, |-\frac{\hbar}{2}\rangle$$



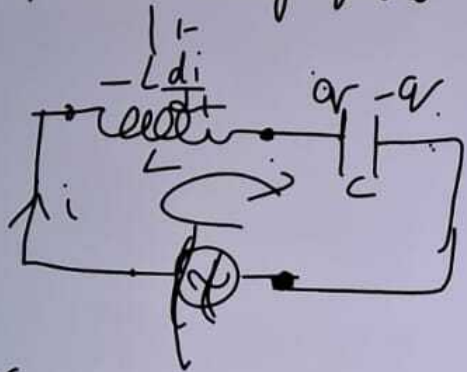
$$\begin{array}{cc} |0\rangle & |1\rangle \\ \updownarrow & \updownarrow \\ |\frac{1}{2}\rangle & |-\frac{1}{2}\rangle \end{array}$$

$|\psi\rangle$ is wavefunction of electron.

A general $|\psi\rangle$ is given by

$$|\psi\rangle = \alpha |\frac{1}{2}\rangle + \beta |-\frac{1}{2}\rangle, \quad \alpha, \beta \in \mathbb{C}$$

③ Superconducting qubit (2 level atom).



LC circuit

$$q = CV$$

$$V = \frac{q}{C}$$

$$KVL = V + V_L + V_C = 0$$

$$-L \frac{di}{dt} - \frac{q}{C} = 0$$

$$V_C = \frac{q}{C}$$

$$= -L \frac{di}{dt} + \frac{q}{C} \Rightarrow$$

$$L \frac{d^2 q}{dt^2} + \frac{q}{C} = 0$$

$$\boxed{-\hbar k u = m \hbar^2 \frac{d^2 \psi}{dx^2}} \quad \psi = \tilde{\psi}_0 e^{i\omega t}$$

$$\omega = \sqrt{\frac{k}{m}} \quad \text{for a mass } m \text{ on a spring } k$$

$$L \frac{d^2 q}{dt^2} = -\frac{q}{C}$$

$$\boxed{\frac{d^2 a}{dt^2} = -\frac{a}{LC}}$$

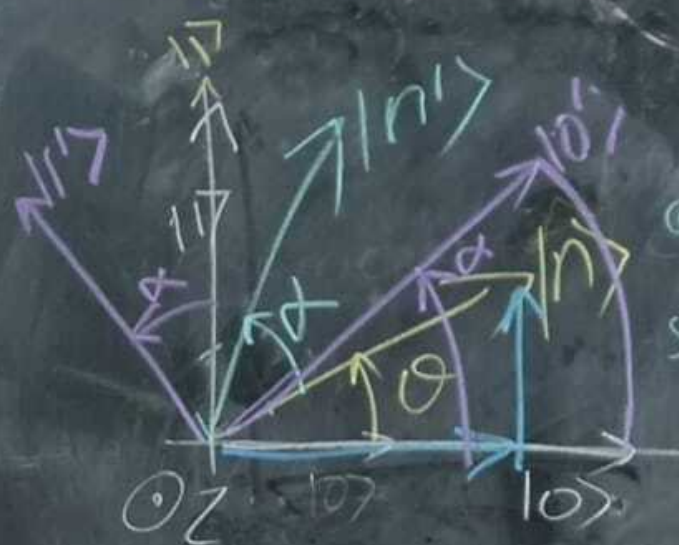
$$q(t) = q_0 \exp(j\omega t)$$

$$\begin{matrix} C & -S & -C \\ S & C & -S \end{matrix} \quad \text{for a mass } m$$

Postulate 2 of QM

Observable postulate

$$\cos\alpha \cos\theta |0\rangle + \cos\alpha \sin\theta |1\rangle - \sin\alpha \sin\theta |0\rangle + \sin\alpha \cos\theta |1\rangle$$



$$|n\rangle = \cos\theta |0\rangle + \sin\theta |1\rangle$$

Method (1)

$$\begin{aligned} & \cos\theta (\cos\alpha |0\rangle + \sin\alpha |1\rangle) \\ & + \sin\theta (-\sin\alpha |0\rangle + \cos\alpha |1\rangle) \\ & = \cos(\alpha + \theta) |0\rangle + \sin(\alpha + \theta) |1\rangle \end{aligned}$$

$$\begin{pmatrix} \cos(\alpha + \theta) \\ \sin(\alpha + \theta) \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

Method (2)
L. Algebra

$$\begin{aligned} & R_2(\alpha) |n\rangle = \\ & \therefore R_2(\alpha) |n\rangle = \\ & |n'\rangle = \\ & R_2(\alpha) [\cos\theta |0\rangle + \sin\theta |1\rangle] \\ & = [\cos\theta R_2(\alpha) |0\rangle + \sin\theta R_2(\alpha) |1\rangle] \end{aligned}$$

$$R_2(\alpha) |0\rangle = |0'\rangle = \cos\alpha |0\rangle + \sin\alpha |1\rangle$$

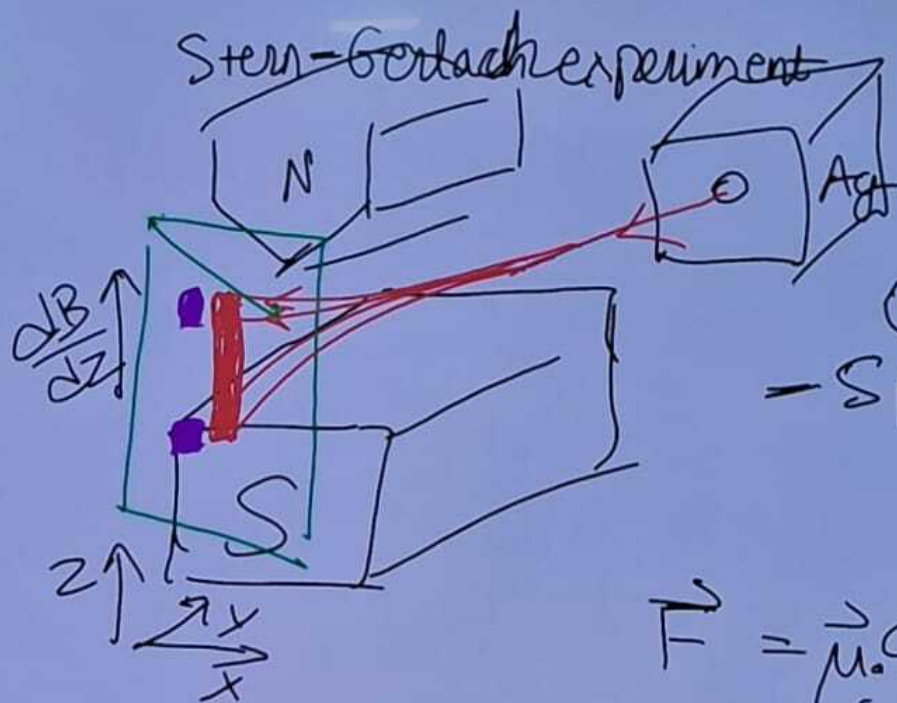
$$R_2(\alpha) |1\rangle = |1'\rangle = -\sin\alpha |0\rangle + \cos\alpha |1\rangle$$

Intermediate step: "Action" of $R_2(\alpha)$ on $\{|0\rangle, |1\rangle\}$

Two level systems

- Polarization states
- Spin $\frac{1}{2}$ system
- Super. Qubit

Observable
 S_x



Observable:
- Spin component
 $E = \vec{\mu} \cdot \vec{B}$

$$\vec{F} = \vec{\mu} \cdot \frac{d\vec{B}}{dz} \propto \mu_z$$

Uhlenbeck - Existence
of spin angular momentum



Postulate 2 of QM.

Observable $\leftrightarrow$ Operator

$$S_z = \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix}$$

Action of S_z

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{cases} S_z \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ S_z \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}$$

(1) If initial state is $|+\frac{1}{2}\rangle$

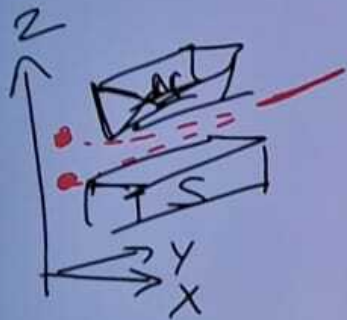
then action of S_z is $\frac{\hbar}{2} |+\frac{1}{2}\rangle$

$$S_z |+\frac{1}{2}\rangle = \frac{\hbar}{2} |+\frac{1}{2}\rangle$$

(2) If initial state is $|-\frac{1}{2}\rangle$

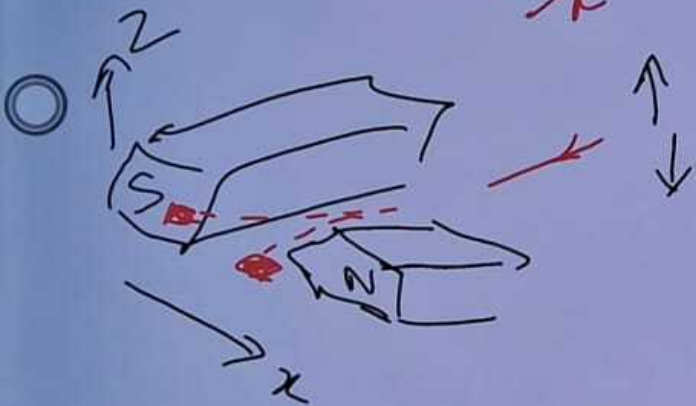
then action of S_z is $-\frac{\hbar}{2} |-\frac{1}{2}\rangle$

$$S_z |-\frac{1}{2}\rangle = -\frac{\hbar}{2} |-\frac{1}{2}\rangle$$



P.T. $|+\frac{1}{2}z\rangle = \frac{1}{\sqrt{2}}(|+\frac{1}{2}z\rangle \pm |-\frac{1}{2}z\rangle)$

HW:
Eigenvalues
& Eigenvectors of
 $\tilde{S}_x$



Action on
x basis
 $\{|+\frac{1}{2}x\rangle, |-\frac{1}{2}x\rangle\}$

(a) $\tilde{S}_x \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ } $\Rightarrow \tilde{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
 (b) $\tilde{S}_x \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ }
 Action of $\tilde{S}_x$ on z basis $\{|+\frac{1}{2}z\rangle\}$

(a) $\tilde{S}_x |+\frac{1}{2}z\rangle = \frac{\hbar}{2} |-\frac{1}{2}z\rangle$ }
 (b) $\tilde{S}_x |-\frac{1}{2}z\rangle = \frac{\hbar}{2} |+\frac{1}{2}z\rangle$ } $\tilde{S}_x =$

$\tilde{S}_x |+\frac{1}{2}x\rangle = +\frac{\hbar}{2} |+\frac{1}{2}x\rangle$
 $\tilde{S}_x |-\frac{1}{2}x\rangle = -\frac{\hbar}{2} |-\frac{1}{2}x\rangle$

Unitary operator:

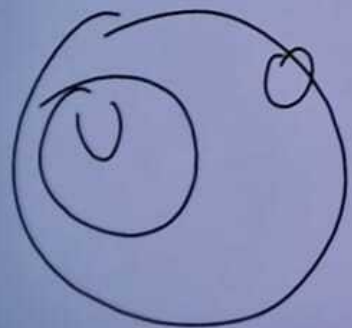
$$\underline{U^\dagger U} = \underline{I} \implies U^{-1} = U^\dagger$$

where $U^\dagger$ is defined as $\Rightarrow$ inverse of unitary always exists and is equal to the conjugate transpose of matrix is equal to the matrix's adjoint

Inverse of matrix.

— A^{-1} is inverse of A
 eff $A^{-1} A = A \cdot A^{-1} = I$
 $\begin{matrix} \nearrow \\ U^\dagger \end{matrix} \begin{matrix} \searrow \\ U \end{matrix} = I$

— The operation of conjugation followed by transpose is called adjoint operation. $\Rightarrow \det U \neq 0$



— The symbol " $\dagger$ " is called dagger.

$$A^{-1} = \frac{\text{adj } A}{\det A}$$

Postulate 3 of QM

Time evolution of a system

CM

- Initial state $\{x(0), p(0)\}$
- Forces
- Final state $\{x(t), p(t)\}$
- $\vec{F}_{net} = \frac{d\vec{p}}{dt}$
- Reversibility?

QM

$|\psi(0)\rangle$

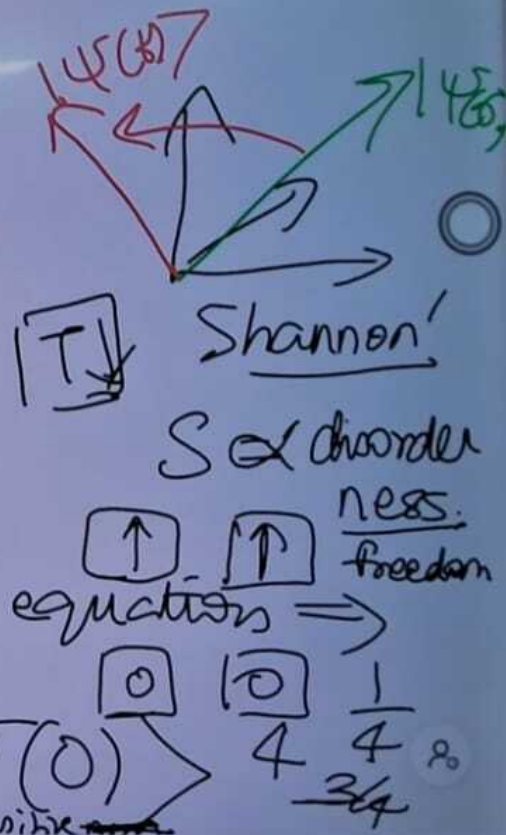
Operators

$|\psi(t)\rangle$

Schrodinger equation $\Rightarrow$

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

Reversible. Unitary



$$| \psi(t) \rangle = U(t) | \psi(0) \rangle$$

Left Multiply o.b.s. by $U^\dagger(t)$

$$U^\dagger | \psi(t) \rangle = \tilde{U}^\dagger(t) \tilde{U}(t) | \psi(0) \rangle$$

$$\text{But } \tilde{U}^\dagger = \tilde{U}^{-1}$$

$$\Rightarrow U^\dagger | \psi(t) \rangle = \tilde{U}^{-1}(t) \tilde{U}(t) | \psi(0) \rangle$$

$$U^\dagger | \psi(t) \rangle = | \psi(0) \rangle$$

Quantum mechanics is perfectly reversible

Postulate 3 of QM

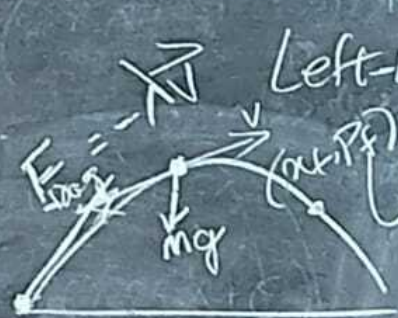
Time evolution postulate

Postulate 3 - QM.

Time evolution of system represented by $|\psi(0)\rangle$ at time $t=0$ is given by unitary matrix $U(t)$

and the final state is given

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle$$



Left-multiply o.b.s. by $U^\dagger(t)$

$$U^\dagger(t)|\psi(t)\rangle = U^\dagger(t)U(t)|\psi(0)\rangle$$

$$\text{But } U^\dagger = U^{-1}$$

$$U^\dagger(t)|\psi(t)\rangle = U^\dagger(t)U(t)|\psi(0)\rangle$$

$$|\psi(0)\rangle = U^\dagger(t)|\psi(t)\rangle$$

Perfect reversibility in QM

Eigenvalues and eigenvectors of the observable $\tilde{S}_n$

— Let us work with the basis vectors $\{|0\rangle, |1\rangle\}$ that are eigenvectors of observable $\tilde{S}_z$

$$\Rightarrow \left. \begin{aligned} \tilde{S}_z |0\rangle &= \frac{\hbar}{2} |0\rangle \\ \tilde{S}_z |1\rangle &= -\frac{\hbar}{2} |1\rangle \end{aligned} \right\} \text{Action of } \tilde{S}_z$$

$$\Rightarrow \tilde{S}_z = \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Action of $\tilde{S}_x$:

$$\tilde{S}_x |0\rangle = \frac{\hbar}{2} |1\rangle$$

$$\tilde{S}_x |1\rangle = \frac{\hbar}{2} |0\rangle$$

$$\tilde{S}_x = \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- Eigenvalues of $\tilde{S}_x$:

$$\det(\tilde{S}_x - \lambda I) = 0 \Leftrightarrow$$

$$\Leftrightarrow \lambda^2 - \left(\frac{\hbar}{2}\right)^2 = 0$$

$$\Leftrightarrow \lambda = \pm \frac{\hbar}{2}$$

Case i) $\lambda = +\frac{\hbar}{2}$ Let $|\frac{\hbar}{2}\rangle = \alpha|0\rangle + \beta|1\rangle$

$$\begin{pmatrix} +\frac{\hbar}{2} & \frac{\hbar}{2} \\ \frac{\hbar}{2} & +\frac{\hbar}{2} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$$

$$\Leftrightarrow +\alpha + \beta = 0$$

$$\Leftrightarrow \alpha = -\beta$$

$$\begin{vmatrix} \frac{\hbar}{2} & -\lambda \\ \frac{\hbar}{2} & -\lambda \end{vmatrix} = 0 \Rightarrow |\frac{\hbar}{2}\rangle = \alpha|0\rangle + \beta|1\rangle$$

But norm of $|\pm \frac{1}{2}_x\rangle = 1$ where $|\pm \frac{1}{2}_x\rangle = \alpha(|0\rangle \pm |1\rangle) = \alpha|0\rangle \pm \alpha|1\rangle$
 $= |\alpha 0 \pm \alpha 1\rangle$

$$\Rightarrow \langle \pm \frac{1}{2}_x | \pm \frac{1}{2}_x \rangle =$$

$$\Leftrightarrow \langle \pm \frac{1}{2}_x | = \langle \alpha 0 \pm \alpha 1 |$$

$$\alpha^* (\langle 0 | \pm \langle 1 |) \cdot \alpha (|0\rangle \pm |1\rangle) = 1$$

$$= \langle \alpha 0 | \pm \langle \alpha 1 |$$

$$= \alpha^* (\langle 0 | \pm \langle 1 |)$$

$$|\pm \frac{1}{2}_x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

$$\left[\begin{array}{l} \langle 0 | 0 \rangle = 1 = \langle 1 | 1 \rangle \\ \langle 0 | 1 \rangle = 0 \end{array} \right]$$

$$\Leftrightarrow |\alpha|^2 (1 + 1) = 1$$

$$\Leftrightarrow |\alpha|^2 = \frac{1}{2} \Leftrightarrow \alpha = \frac{1}{\sqrt{2}}$$

$$\therefore |\pm \frac{1}{2}_x\rangle = \frac{1}{\sqrt{2}} |0\rangle \pm \frac{1}{\sqrt{2}} |1\rangle$$

$$\therefore |\pm \frac{1}{2}_x\rangle = \frac{1}{\sqrt{2}} [|0\rangle \pm |1\rangle]$$

Light
 $|H\rangle, |V\rangle$

$$|L_{\pm\frac{\pi}{4}}\rangle$$

$$= |C_{\pm}\rangle$$



$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm j \end{pmatrix}$$

$$|L_{\theta}\rangle = \cos\theta |H\rangle + \sin\theta |V\rangle$$

Spin $\frac{1}{2}$

$\tilde{S}_z$

$$\left\{ \left| \pm \frac{1}{2} \right\rangle_z \right\}$$

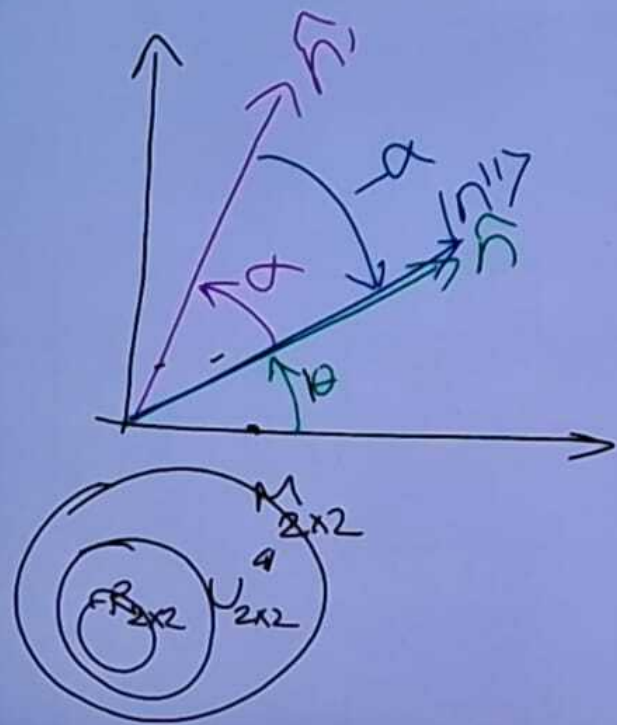
$\tilde{S}_x$

$$\left\{ \left| \pm \frac{1}{2} \right\rangle_x \right\}$$

$\tilde{S}_y$

$$\left\{ \left| \pm \frac{1}{2} \right\rangle_y \right\}$$

Are the rotation matrices $R_Z(\alpha)$ unitary?



$$R_Z(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$R_Z(-\alpha) = \begin{pmatrix} \cos \alpha & +\sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$$R_Z^\dagger(\alpha) \stackrel{?}{=} R_Z^{-1}$$

$$R^{-1} = \frac{1}{\det R} \text{adj} R$$

$$(R_Z^*(\alpha))^T = R_Z^T(\alpha) \stackrel{?}{=} R_Z^{-1}$$

$$\Leftrightarrow R_Z^{-1}(\alpha) = R_Z(-\alpha)$$

$$R^{-1} R = I$$

$$\Leftrightarrow R_Z^T(R|\psi\rangle) = I|\psi\rangle$$

$$\Leftrightarrow R_Z^{-1}|n\rangle = |\psi\rangle$$

Postulate 4 - QM

Measurement $\checkmark$ Born rule by Max Born (Probabilistic interpretation of wavefunctions)

(a) Given an observable $\hat{A}$ with eigenvalues $\{a_i\}$ and corresponding eigenvectors $\{|a_i\rangle\}$, $i = \{1, 2, \dots, d\}$,

a measurement of observable A on state $|\psi\rangle$ gives

an "outcome" a_i with probability $P(i) = |\langle \psi | a_i \rangle|^2$

(b) Post-measurement state

If outcome a_i is obtained, then the initial state $|\psi\rangle$ "collapses" to the corresponding eigenstate $|a_i\rangle$.

Unitary transformations

Active and Passive transformations

$\tilde{S}_x$ Action: (i) $\tilde{S}_x |0\rangle = \frac{\hbar}{2} |1\rangle$

$$\tilde{S}_x = \frac{\hbar}{2} (|0\rangle\langle 1| + |1\rangle\langle 0|) |0\rangle$$

$$\parallel \underbrace{\frac{\hbar}{2} (|0\rangle\langle 1|)}_{\text{outer ket}} |0\rangle + \frac{\hbar}{2} (|1\rangle\langle 0|) |0\rangle$$

Decomposition

$$a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= a |0\rangle\langle 0| + b |0\rangle\langle 1| + c |1\rangle\langle 0| + d |1\rangle\langle 1|$$

$$= \frac{\hbar}{2} |0\rangle (\underbrace{\langle 1|0\rangle}_{\text{ket-inner}}) + \frac{\hbar}{2} |1\rangle (\langle 0|0\rangle)$$

$$= \frac{\hbar}{2} |1\rangle$$

(ii)

$$\tilde{S}_x |1\rangle$$

$$= \frac{\hbar}{2} |0\rangle$$

$$\tilde{S}_x = \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix}$$

$$= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \frac{\hbar}{2} (|1\rangle\langle 0| + |0\rangle\langle 1|)$$

Operator in Dirac notation

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}_{2 \times 2}$$

 $a, b, c, d \in \mathbb{C}$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \frac{\hbar}{2} \left[\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \right]$$

$$= \frac{\hbar}{2} (|1\rangle\langle 0| + |0\rangle\langle 0|)$$

$$= \frac{\hbar}{2} (|1\rangle\langle 0| + |0\rangle\langle 1|)$$

H.W.

Express eigenvectors of $\tilde{S}_z$
in terms of
eigenvectors of $\tilde{S}_x$

$$|+\frac{1}{2}_z\rangle = \alpha |+\frac{1}{2}_x\rangle + \beta |-\frac{1}{2}_x\rangle$$
$$= \alpha |+\rangle + \beta |-\rangle$$

(i) + (ii) gives $|+\rangle + |-\rangle = \frac{\sqrt{2}}{2} |0\rangle$

$$|0\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle)$$

Method A
Mathematician's way

(i) - (ii) gives

$$|+\rangle - |-\rangle = \sqrt{2} |1\rangle$$
$$\Rightarrow |1\rangle = \frac{1}{\sqrt{2}} (|+\rangle - |-\rangle)$$

We expressed

eigenvectors of $\tilde{S}_x$

in terms of
eigenvectors of $\tilde{S}_z$

$$|+\frac{1}{2}_x\rangle = \frac{1}{\sqrt{2}} (|+\frac{1}{2}_z\rangle + |-\frac{1}{2}_z\rangle)$$
$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

i.e. $|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$ (i)

$|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$ (ii)

Method (B)

Physics point of view

$$\tilde{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

eigenvalues = $+\frac{\hbar}{2}$

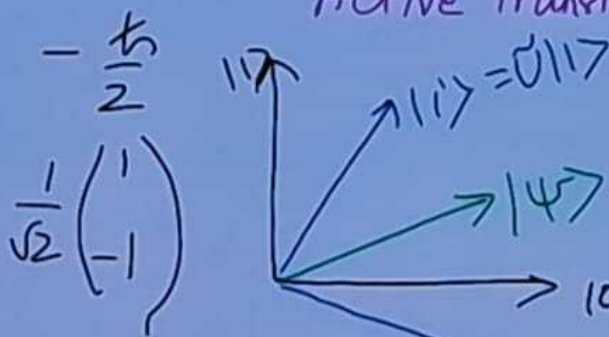
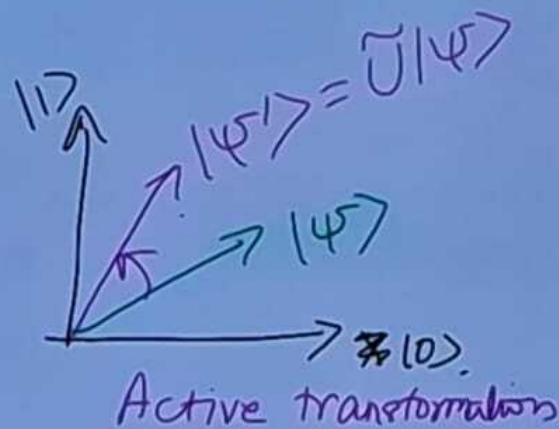
eigenvectors = $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Let $\tilde{U} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$

so that the diagonal matrix

$$\tilde{U}^\dagger \tilde{S}_x \tilde{U}$$

is given by



$$\tilde{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\tilde{S}_z |0\rangle = \frac{\hbar}{2} |0\rangle$$

$$\tilde{S}_z |1\rangle = -\frac{\hbar}{2} |1\rangle$$

$$\tilde{A} \vec{x} = \lambda \vec{x}$$

Passive transformations

$$\tilde{U}^+ |+\rangle = \tilde{U}^+ |0\rangle = |0\rangle$$

$$\tilde{U}^+ |-\rangle = \tilde{U}^+ |1\rangle = |1\rangle$$

$$\tilde{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\tilde{S}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$|\frac{1}{2}_x\rangle = |+\rangle \equiv |0'\rangle = \tilde{U} |0\rangle = \tilde{U} |+\frac{1}{2}_z\rangle$$

$$|-\frac{1}{2}_x\rangle = |-\rangle \equiv |1'\rangle = \tilde{U} |1\rangle = \tilde{U} |-\frac{1}{2}_z\rangle$$

$$\tilde{U}^+ \tilde{S}_x U = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & +1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

But we know

$$\tilde{U}^+ = \tilde{U}^-$$

so that if we left-multiply
obs. by $\tilde{U}^+$, then

Now,
eigenbasis is $|+\rangle$ and $|-\rangle$
so that

$\tilde{U}$ in the new basis is

$$\tilde{U}^\dagger \tilde{U} \tilde{U} = \tilde{U}^\dagger \tilde{U} \tilde{U} = \tilde{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Therefore $|0\rangle = \tilde{U}^\dagger |+\rangle$

only

$$|1\rangle = \tilde{U}^\dagger |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} (|+\rangle - |-\rangle)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

only

$$\frac{1}{\sqrt{2}} (|+\rangle + |-\rangle)$$

$$\therefore |0\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle)$$

Postulate 4 of QM

Measurement postulate

P I

Wavefunction

— $|\psi\rangle \in \mathcal{H}$

Rules of linear vector spaces are applicable

— $\langle \psi | \psi \rangle = 1$

— $|\psi\rangle$ are column vectors

— $\langle \psi |$, dual of $|\psi\rangle$ are row vectors

P II

Operator/Observable

— $\tilde{O} |\psi\rangle = |\psi'\rangle$

Operators "act on"

kets and transform them to another ket

$= \langle \psi(0) | \psi(0) \rangle$

— Operators are Hermitian matrices . i.e.

$\langle \psi(t) | \psi(t) \rangle = \langle \psi(0) | \psi(0) \rangle$

P III

Time evolution

$U(t) |\psi(0)\rangle$

$\langle \psi(0) | U^\dagger(t) U(t) | \psi(0) \rangle = \langle \psi(0) | \psi(0) \rangle$

Here, evolution operator is unitary.

— So that norm is conserved.

$\langle U^\dagger(t) \psi(0) | U(t) \psi(0) \rangle$

$|\psi(0)\rangle = U^\dagger |\psi(t)\rangle$

$U^\dagger U |\psi(0)\rangle = U^\dagger |\psi(t)\rangle$

P IV

Measurement: Given observable/operator $\tilde{A}$ with eigenvalues $\{a_i\}$ and

$$\langle a_j | [\psi] \rangle = \sum_i c_i \langle a_j | a_i \rangle \Rightarrow \langle a_j | \psi \rangle = \sum_i c_i \langle a_j | a_i \rangle$$

eigenvectors $\{|a_i\rangle\}$

(i) then a measurement of observable will lead to an outcome a_i with a probability of $|c_i|^2 = \langle a_i | \psi \rangle^2$

(ii) ~~After~~ ^{If} the measurement ^{of outcome a_i} then system "collapses" ^{the} to the eigenstate $|a_i\rangle$.

Sequential measurements

P. 327 (G)

$$\tilde{A} \leftrightarrow \{|\psi_1\rangle, |\psi_2\rangle\} \text{ with } \{a_1, a_2\}$$

$$\tilde{B} \leftrightarrow \{|\phi_1\rangle, |\phi_2\rangle\} \text{ with } \{b_1, b_2\}$$

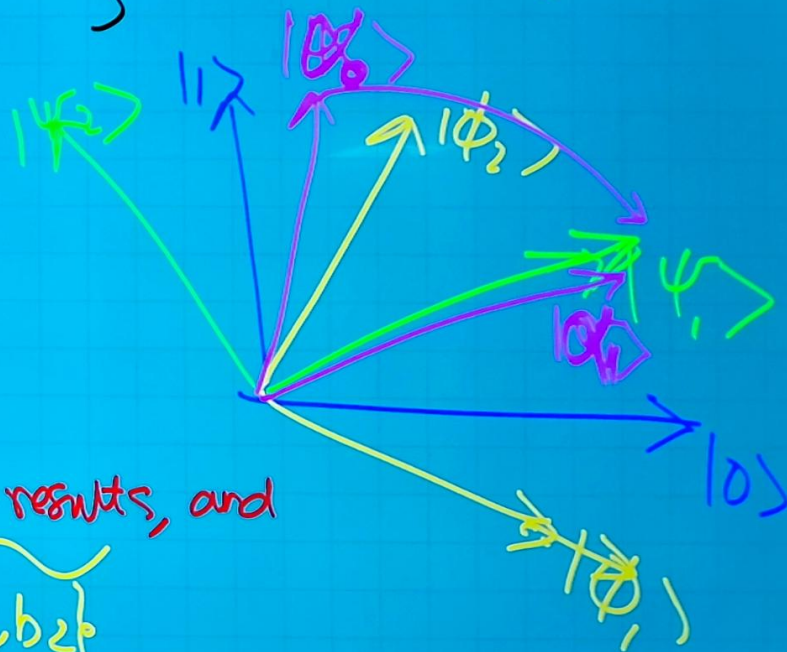
(*) Observer A is measured, and outcome a_1 is obtained.

What is the state of the system after this measurement?

Now.

$$|\psi_1\rangle = \frac{1}{5}[3|\phi_1\rangle + 4|\phi_2\rangle]$$
$$|\psi_2\rangle = \frac{1}{5}[4|\phi_1\rangle - 3|\phi_2\rangle]$$

(b) If B is now measured, what are the possible results, and what are their probabilities?



$\{b_1, b_2\}$

$$|\alpha_1\rangle = |\psi_1\rangle = \frac{1}{5} [3|\phi_1\rangle + 4|\phi_2\rangle]$$

Reality check
 $\langle \psi_1 | \psi_1 \rangle = 1$

B

:

Outcome

probability

b_1

$$|\langle \phi_1 | \psi_1 \rangle|^2$$

$$= \left| \langle \phi_1 | \frac{1}{5} [3|\phi_1\rangle + 4|\phi_2\rangle] \right|^2 = \frac{1}{25} [16 + 9]$$

$$= \left| \frac{3}{5} \right|^2 = \frac{9}{25}$$

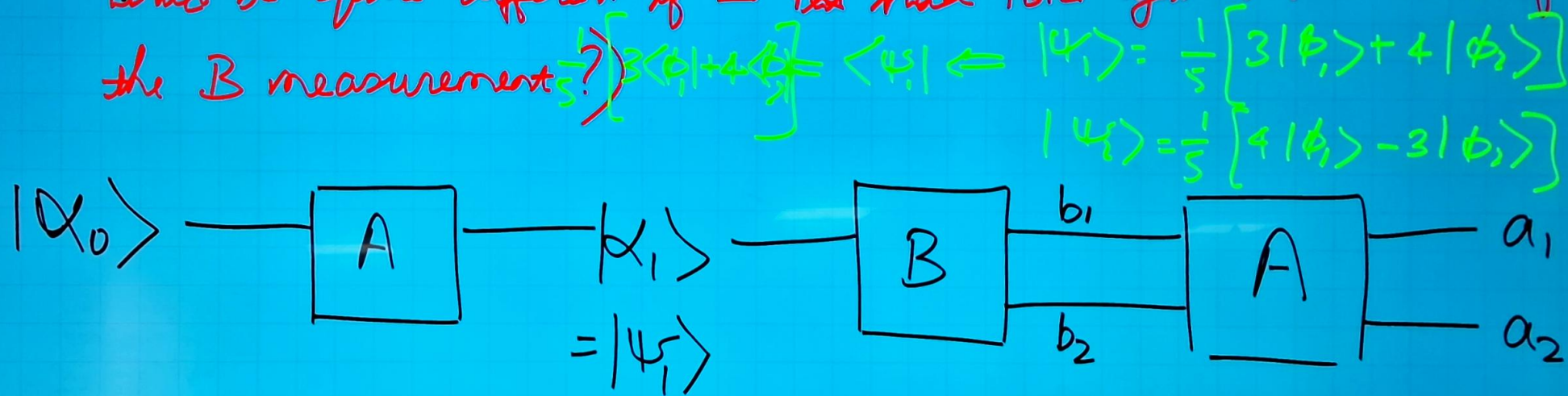
b_2

$$|\langle \phi_2 | \psi_1 \rangle|^2 = \frac{16}{25}$$

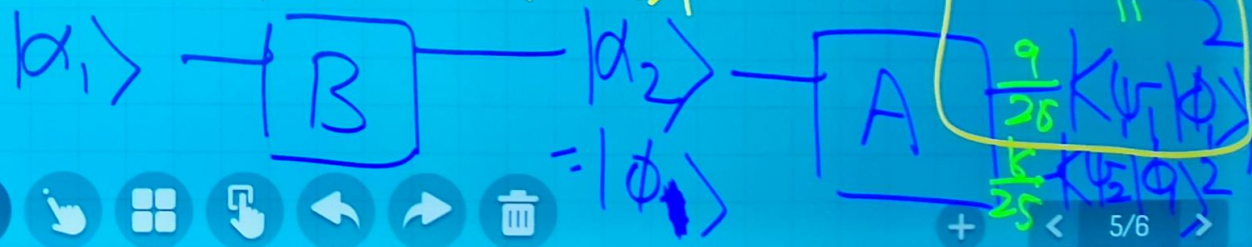
$$\frac{1}{5} [3\langle \phi_1 | + 4\langle \phi_2 |]$$

$$\frac{1}{5} [3|\phi_1\rangle + 4|\phi_2\rangle]$$

(c) Right after the measurement of B, A is measured again. What is the probability of getting a_1 ? (Note that the answer could be quite different if I had told you the outcome of the B measurement.)



Case (i) Post-B, the outcome is b_1 . $\frac{9}{25} = |\langle \phi_1 | \psi_1 \rangle|^2$



Case (ii), Post-B, the outcome is b_2

$$\alpha_1 \xrightarrow{B} \alpha_2 \xrightarrow{A} \alpha_3$$

$\alpha_1 = |\psi_1\rangle$
 $\alpha_2 = |\phi_2\rangle$
 $\alpha_3 = |\psi_1\rangle$

$\frac{16}{25} = |\langle \phi_2 | \psi_1 \rangle|^2$
 $\frac{9}{25} = |\langle \psi_2 | \phi_2 \rangle|^2$

Probability of $|\alpha_3\rangle$ being $|\psi_1\rangle$, given that $|\alpha_2\rangle$ is $|\phi_2\rangle$ is $\frac{9}{25}$

Prob of $|\alpha_3\rangle$ being $|\psi_1\rangle$, given that $|\alpha_2\rangle$ is $|\phi_1\rangle$ is $\frac{16}{25}$

Then, the prob of $|\alpha_3\rangle$ being $|\psi_1\rangle$ is

$$0.5 > = \frac{337}{625} = \frac{9}{25} \cdot \frac{9}{25} + \frac{16}{25} \cdot \frac{16}{25}$$

Eigenvalue and expectation value

In Dirac Notation

Expectation value of observable Q with $Q|i\rangle = q_i|i\rangle$ $i \in \{1, 2, \dots, d\}$

Def: Weighted average of outcome of operator/observable

— If $|\psi\rangle = \sum_{i=1}^d c_i |i\rangle$, where $c_i = \langle i|\psi\rangle$

then $\langle Q \rangle := \sum_{i=1}^d |c_i|^2 q_i$

$$\tilde{Q} = \begin{pmatrix} q_0 & 0 & 0 & \dots & 0 \\ 0 & q_1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & q_d \end{pmatrix} =$$

Aim: To rewrite expectation value $\bar{c}$ Bracket notation

$$\langle Q \rangle = \langle \psi | \sum_{i=1}^d q_i |i\rangle \langle i| \psi \rangle = \sum_{i=1}^d |c_i|^2 q_i = \sum_{i=1}^d \langle \psi | i \rangle q_i \langle i | \psi \rangle$$

$|c_i|^2 = c_i^* c_i$

$\langle Q \rangle = \langle \psi | \tilde{Q} | \psi \rangle$

$\left. \begin{aligned} & q_0 |0\rangle \langle 0| \\ & + q_1 |1\rangle \langle 1| \\ & + \dots \\ & + q_d |d\rangle \langle d| \end{aligned} \right\} = \sum_{i=1}^d q_i |i\rangle \langle i|$

$\langle \tilde{Q} \rangle$ must be real. $\approx$

$\Rightarrow$ any eigenvalue of observable must be real.

Question: When eigenvalue of operator is real always?

Given $\tilde{Q} |i\rangle = q_i |i\rangle$

$$Q_{ij} = \tilde{Q} = \begin{pmatrix} q_0 & 0 & 0 & \dots & 0 \\ 0 & q_1 & 0 & \dots & 0 \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & q_d \end{pmatrix} = \sum_{i=1}^d q_i |i\rangle \langle i|$$

$\langle i | \tilde{Q} | i \rangle$

$$\rightarrow \begin{matrix} j=0 \\ \begin{pmatrix} q_0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \end{matrix} = \tilde{Q} |0\rangle$$

$$\Leftarrow \langle 0 | \tilde{Q} | 0 \rangle = q_0 = (1 \ 0 \ 0 \ \dots \ 0) \begin{pmatrix} q_0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Left multiplying $\tilde{Q}|i\rangle = q_i|i\rangle$ with $\langle j|$ o.b.s.

$$\langle j|\tilde{Q}|i\rangle = q_i\langle j|i\rangle$$

Taking complex conjugate on b.s.

$$\langle j|\tilde{Q}|i\rangle^* = (q_i\langle j|i\rangle)^* = q_i^*\langle j|i\rangle^*$$

$$\begin{aligned} \hookrightarrow \langle j|\tilde{Q}|i\rangle^* &= \langle \tilde{Q}i|j\rangle = q_i\langle i|j\rangle \\ &= \langle i|\tilde{Q}^\dagger|j\rangle = q_i\langle i|j\rangle = \langle i|\tilde{Q}|j\rangle \end{aligned}$$

i.e. If $\text{adj}(Q) = \tilde{Q}^\dagger$, then physical operator must satisfy
adjoint of $Q = \tilde{Q}$
i.e. Q must be Hermitian

$$\boxed{\tilde{Q}^\dagger = \tilde{Q}}$$

$$|\psi\rangle \rightarrow |C\psi\rangle = C|\psi\rangle$$

$$\langle\psi| \rightarrow \langle C\psi| = C^*\langle\psi|$$

$$Q^\dagger = \begin{pmatrix} q_1 & 0 & 0 \\ 0 & q_2 & 0 \\ 0 & 0 & \ddots \end{pmatrix}$$



General properties of Hermitian operators

In computational basis $\{|1\rangle, |0\rangle\}$,

observable $\tilde{Q}$ is not diagonal i.e.

$$\text{lets say } \tilde{Q} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\Rightarrow \tilde{Q} = \begin{pmatrix} \alpha & \gamma \\ \gamma^* & \beta \end{pmatrix}$$

$$\alpha^* = \alpha \\ \beta^* = \beta$$

$$c^* = b \\ b^* = c$$

where α, β are real, $\gamma \in \mathbb{C}$

Write down conditions satisfied by q_{ij} .

Since $\tilde{Q}$ is Hermitian, (Hermitian)

$$\tilde{Q}^\dagger = \tilde{Q}$$

$$\Leftrightarrow (\tilde{Q}^*)^T = \tilde{Q} \Leftrightarrow$$

$$\begin{pmatrix} \alpha^* & \gamma^* \\ c^* & \beta^* \end{pmatrix}^T = \begin{pmatrix} \alpha & b \\ c & d \end{pmatrix}$$

$$\begin{pmatrix} \alpha^* & c^* \\ b^* & \beta^* \end{pmatrix} = \begin{pmatrix} \alpha & b \\ c & d \end{pmatrix}$$

Completeness relation

Express $\tilde{I}$ in Dirac notation

Action of $\tilde{I}$ on computational basis

$$\tilde{I}|i\rangle = |i\rangle$$

$$\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & 0 \\ 0 & 0 & \ddots & 0 \\ \vdots & \vdots & & \ddots \\ 0 & 0 & & 1 \end{pmatrix}$$

$$\begin{aligned} &= \sum_i \langle \psi | i \rangle \langle i | \psi \rangle \\ &= \sum_i c_i^* c_i = \sum_i |c_i|^2 \end{aligned}$$

Method ①

$$\langle \psi | \psi \rangle$$

$$= \langle \psi | \tilde{I} \psi \rangle$$

$$= \langle \psi | \tilde{I} | \psi \rangle$$

$$= \langle \psi | \sum_i |i\rangle \langle i| \psi \rangle$$

$$= |0\rangle \langle 0| + |1\rangle \langle 1| + \dots + |d\rangle \langle d|$$

$$= \sum_{i=1}^d |i\rangle \langle i| = \tilde{I}$$

Power of completeness relation

In $\{|i\rangle\}$ computational basis,

if $|\psi\rangle = \sum_{i=1}^d c_i |i\rangle$, then

find the norm of ψ .

Method ②

$$\langle \psi | \psi \rangle = \langle \psi | \sum_i c_i |i\rangle$$

$$= \left\langle \sum_i c_i |i\rangle \left| \sum_j c_j |j\rangle \right. \right\rangle$$

$$= \sum_i \sum_j c_i^* \langle i | j \rangle c_j$$

$$= \sum_i |c_i|^2 = \langle \psi | \psi \rangle$$

Write Dirac notation for

Hermitian operator.

$$\tilde{Q} = \begin{pmatrix} \alpha & \gamma \\ \gamma^* & \beta \end{pmatrix}, \begin{matrix} \alpha, \beta \in \mathbb{R} \\ \gamma \in \mathbb{C} \end{matrix}$$



Postulates of QM

(1) Wavefunction postulate

Wavefn live in Hilbert space

- $|\psi\rangle \in \mathcal{H}$ Hilbert space is a
d-dimensional, complex
valued field, linear vector space

- $\langle\psi|$ is the dual of $|\psi\rangle$

- $\langle\psi|\psi\rangle = 1$ and $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$

- Matrix representation of $|\psi\rangle$ is d-dimensional
column vector

- If $\{|i\rangle\}$ is the computational basis, then $|\psi\rangle = \sum_i c_i |i\rangle$, where $c_i = \langle i|\psi\rangle$

(2) Observable postulate

Observables are Hermitian operators

- An observable $q(x,p)$ is associated with an operator $\hat{Q}$ such that its expectation value is real.

$$\hat{Q} = \begin{pmatrix} q_0 & 0 \\ 0 & q_1 \end{pmatrix} = q_0 |0\rangle\langle 0| + q_1 |1\rangle\langle 1|$$

Hermitian operator

1. Completeness relation
2. Uncertainty of observable
3. Deterministic state

$$\tilde{Q} = \begin{pmatrix} q_0 & 0 \\ 0 & q_1 \end{pmatrix}$$

$$= q_0 |0\rangle\langle 0| + q_1 |1\rangle\langle 1| = \sum_{i=0}^1 q_i |i\rangle\langle i|$$

2D Hermitian operator $\left(\begin{matrix} \langle 0|\tilde{Q}|0\rangle & \langle 0|\tilde{Q}|1\rangle \\ \langle 1|\tilde{Q}|0\rangle & \langle 1|\tilde{Q}|1\rangle \end{matrix} \right)$ when $\{ \text{computational basis} \} = \{ \text{eigen basis} \}$

$$\tilde{Q} = \begin{pmatrix} \alpha & \gamma \\ \gamma^* & \beta \end{pmatrix}, \alpha, \beta \in \mathbb{R}, \gamma \in \mathbb{C}$$

Here $\{ \text{computational basis} \}$

$$\tilde{Q} = \begin{pmatrix} q_0 & 0 \\ 0 & q_1 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \cdot (1 \ 0) \neq \{ \text{eigenbasis basis} \}$$

$$= q_0 |0\rangle\langle 0| + q_1 |1\rangle\langle 1| = \sum_{i=0}^1 q_i |i\rangle\langle i|$$

$$= \alpha |0\rangle\langle 0| + \beta |1\rangle\langle 1| + \gamma |0\rangle\langle 1| + \gamma^* |1\rangle\langle 0|$$

$$\tilde{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix}$$

← Here, computational basis is $\{|0\rangle, |1\rangle\} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

(1) Q: Is $\tilde{S}_x$ Hermitian?

$$\tilde{S}_x \text{ is Hermitian} \Leftrightarrow \tilde{S}_x^\dagger = \tilde{S}_x$$

$$\Leftrightarrow (\tilde{S}_x^*)^T = \tilde{S}_x \Leftrightarrow \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix}^T = \begin{pmatrix} 0 & \frac{\hbar}{2} \\ \frac{\hbar}{2} & 0 \end{pmatrix} \Rightarrow \text{true statement}$$

(2) Q: If yes, what is its Dirac notation? Obtain in two ways

(i) Mathematicians (ii) Physicists

$$(i) \quad \tilde{S}_x = \frac{\hbar}{2} (|0\rangle\langle 1| + |1\rangle\langle 0|)$$

(ii) If $\{\text{computational basis}\} = \{\text{eigenbasis of } \tilde{S}_x\}$

$$\text{Then } \tilde{S}_x = \sum_{i=0}^1 S_i^x |i\rangle\langle i|$$

(ii) But $\tilde{S}_x$ is not diagonal in the computational basis.

$$\tilde{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Eigen values & Eigenkets.

$$\lambda = +\frac{\hbar}{2}$$

$$|+\frac{1}{2}_x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\lambda = -\frac{\hbar}{2}$$

$$|-\frac{1}{2}_x\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Therefore, Dirac notation for $\tilde{S}_x = +\frac{\hbar}{2} |+\frac{1}{2}_x\rangle \langle +\frac{1}{2}_x| - \frac{\hbar}{2} |-\frac{1}{2}_x\rangle \langle -\frac{1}{2}_x|$

$$\tilde{S}_x = \frac{\hbar}{2} \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right] \left[\frac{1}{\sqrt{2}}(\langle 0| + \langle 1|) \right] - \frac{\hbar}{2} \left[\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right] \left[\frac{1}{\sqrt{2}}(\langle 0| - \langle 1|) \right]$$

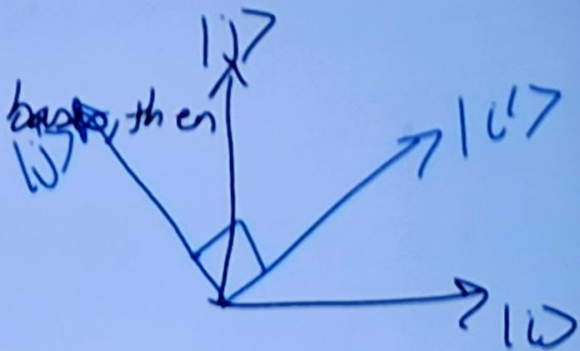
$$= \frac{\hbar}{2} \cdot \frac{1}{2} [2|0\rangle\langle 1| + 2|1\rangle\langle 0|]$$

$$= \frac{\hbar}{2} \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right] \left[\frac{1}{\sqrt{2}}(\langle 0| + \langle 1|) \right] = \frac{\hbar}{2} (|0\rangle\langle 1| + |1\rangle\langle 0|)$$



Completeness relation - If $\{|i\rangle\}$ is the computational basis, then

$$\underline{I} = \sum_{i=0}^{d-1} |i\rangle\langle i|$$



Now, understandably, computational basis spans the ⁶⁶complete ⁹⁹vector space

So computational basis is a complete basis i.e. $|\psi\rangle = \sum_{i=0}^{d-1} c_i |i\rangle$

- Now, $\{\text{eigenfunctions/eigenkets}\}$ of Hermitian operator $\tilde{Q}$ also forms a ^(Axiom) $\rightarrow$ complete basis.
i.e. for $\{q_i\}$ eigenvalues with $\{|q_i\rangle\}$, $|\psi\rangle = \sum_{i=0}^{d-1} \alpha_i |q_i\rangle$



...

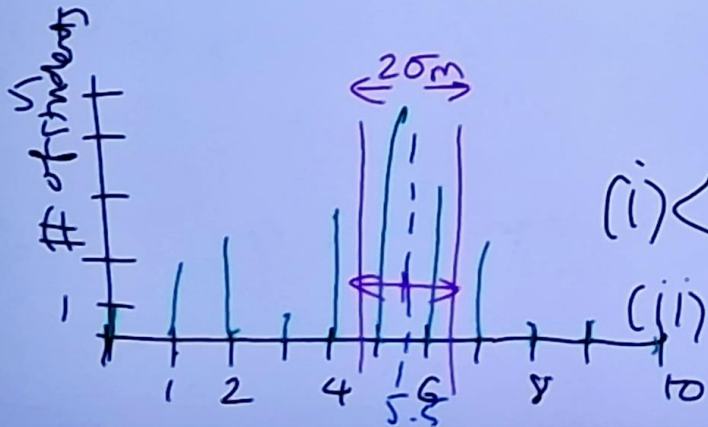


4/9

Uncertainty of observable (σ)

Given an observable Q , the uncertainty of Q is defined as expectation value of root mean square of deviation

$$\sigma_Q = \sqrt{\langle (Q - \langle Q \rangle)^2 \rangle}$$



$$(i) \langle m \rangle = \frac{\sum_{i=1}^N m_i}{N} \quad (iii) \langle d^2 \rangle = \frac{\sum_{i=1}^N d_i^2}{N}$$

$$(ii) d_i = m_i - \langle m \rangle$$

$$m = \langle \bar{m} \rangle \pm \sigma_m = 5.5 \pm 1$$

$$(iv) \sigma_m = \sqrt{\langle d^2 \rangle} = \sqrt{\frac{\sum (m_i - \langle m \rangle)^2}{N}}$$

Problem

Uncertainty in $\tilde{S}_x$

Given $|\psi\rangle = |0\rangle$

find uncertainty in $\tilde{S}_x$ i.e. σ_{S_x}

$$\sigma_{S_x}^2 = \langle (\tilde{S}_x - \langle S_x \rangle)^2 \rangle$$

(ii) Expectation value of $\tilde{S}_x^2$ is

$$\langle \tilde{S}_x^2 \rangle$$

$$\begin{aligned} \text{But } \tilde{S}_x^2 &= \tilde{S}_x \tilde{S}_x \\ &= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

$\therefore \langle \tilde{S}_x \rangle =$ Expectation value of $Q := \langle Q \rangle = \langle \psi | \tilde{Q} | \psi \rangle$

$\begin{matrix} 0 \\ 1 \end{matrix}$ (i) Expectation value of S_x is $\langle S_x \rangle = \langle \psi | \tilde{S}_x | \psi \rangle$

$$\frac{\hbar}{2} \langle 0 | 1 \rangle = \langle 0 | \frac{\hbar}{2} | 1 \rangle \langle 0 | 0 \rangle + \langle 0 | \frac{\hbar}{2} | 0 \rangle \langle 1 | 0 \rangle$$

$$\leftarrow \langle 0 | \frac{\hbar}{2} (|0\rangle\langle 1| + |1\rangle\langle 0|) | 0 \rangle$$



$$\tilde{S}_x^2 = \frac{\hbar^2}{4} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\tilde{S}_x^2 = \frac{\hbar^2}{4} \mathbb{I}_{2 \times 2}$$

$$\therefore \langle \tilde{S}_x^2 \rangle = \langle \Psi | \tilde{S}_x^2 | \Psi \rangle$$

$$= \langle 0 | \frac{\hbar^2}{4} \mathbb{I} | 0 \rangle$$

$$= \frac{\hbar^2}{4} \langle 0 | 0 \rangle = \frac{\hbar^2}{4} \langle \tilde{S}_x^2 \rangle$$

$$\therefore \sigma_{S_x} = \sqrt{\langle S_x^2 \rangle}$$

$$= \sqrt{\frac{\hbar^2}{4}}$$

$$\boxed{\sigma_{S_x} = \frac{\hbar}{2}}$$



Deterministic state.

State with zero uncertainty in observable $\tilde{Q}$ is called deterministic state with respect to $\tilde{Q}$.

Claim: Eigenket of $\tilde{Q}$ is deterministic state of $\tilde{Q}$

Reason: Given $|q_i\rangle$ is eigenket of $\tilde{Q}$ with eigenvalue q_i

then $\tilde{Q} |q_i\rangle = q_i |q_i\rangle \Rightarrow$ Again a measurement of $\tilde{Q}$ is done

Stationary state

Deterministic state of Hamiltonian operator is called stationary state.

Energy/Hamiltonian operator $\tilde{H} = \tilde{T} + \tilde{V}$

